CS PhD Seminar Series

November 10th | 16:00-17:00 | Room 322

An Introduction to Linear Logic

Classical mathematical logic models permanent truth: an assumption can be duplicated or discarded in writing a proof. Introduced by Girard in 1987, linear logic reframes this perspective, by treating assumptions as finite resources rather than permanent truths. This means that under linear constraints, any assumption has to be used exactly once in a proof. This logic finds practical applications in computer science, such as tracking file handlings safely, managing concurrent systems, and modelling automatic differentiation.



Speaker: **Andrea Giusto**

Andrea Giusto is a PhD student in Computer Science at University of Genoa. His research lies at the intersection of type theory and category theory, and he is focusing on a project aiming to formalize Automatic Differentiation on manifolds. He obtained his Bachelor and Master's degrees in Mathematics at University of Genoa, studying categorical models of dependent type theory.

Learning Under Privileged Information in Deep Neural Networks

In many real-world applications, the information available during training is richer than what a model will have access to once deployed: additional sensing modalities, costly measurements, or expert annotations that cannot be collected at test time. Learning Under Privileged Information (LUPI) is the paradigm that studies how such training-only knowledge can be exploited to build better models that rely on standard inputs alone. Originally formulated for kernel methods, the paradigm has increasingly been adopted in deep learning, where privileged information can shape internal representations, guide the optimization process, or be transferred from teacher to student networks. This seminar will introduce the LUPI framework and give an overview of how privileged information has been integrated into deep learning models. Particular attention will be devoted to a question that remains largely open: when does privileged information actually help, and when does it fail to make a difference. The talk will conclude with a brief look at open challenges and recent directions in the field.

Speaker: **Veronica Pignedoli**

Veronica Pignedoli is a PhD student in Computer Science at the Machine Learning Genoa Center (MaLGa), University of Genoa, where she previously earned her Bachelor's degree in Mathematics and her Master's degree in Applied Mathematics. Her research lies at the intersection of machine learning and computer vision, with a focus on cross-modal knowledge transfer and multimodal learning in biomedical imaging.

